Part 1: Linear Algebra (20 points)

Q1 (5 points)

Given

$$A = \begin{bmatrix} 3 & 4 \\ 2 & 1 \end{bmatrix}, \qquad B = \begin{bmatrix} 1 & 2 \\ 0 & -1 \end{bmatrix}.$$

(a) AB

$$\begin{aligned} AB &= \begin{bmatrix} 3 & 4 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 0 & -1 \end{bmatrix} \\ &= \begin{bmatrix} 3(1)+4(0) & 3(2)+4(-1) \\ 2(1)+1(0) & 2(2)+1(-1) \end{bmatrix} \\ &= \begin{bmatrix} 3 & 2 \\ 2 & 3 \end{bmatrix}. \end{aligned}$$

(b) BA

$$\begin{aligned} BA &= \begin{bmatrix} 1 & 2 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 3 & 4 \\ 2 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 1(3)+2(2) & 1(4)+2(1) \\ 0(3)+(-1)(2) & 0(4)+(-1)(1) \end{bmatrix} \\ &= \begin{bmatrix} 7 & 6 \\ -2 & -1 \end{bmatrix}. \end{aligned}$$

(c) $(A+B)^T$

$$\begin{aligned} A + B &= \begin{bmatrix} 3 & 4 \\ 2 & 1 \end{bmatrix} + \begin{bmatrix} 1 & 2 \\ 0 & -1 \end{bmatrix} \\ &= \begin{bmatrix} 4 & 6 \\ 2 & 0 \end{bmatrix}. \end{aligned}$$

$$(A+B)^T = \begin{bmatrix} 4 & 2 \\ 6 & 0 \end{bmatrix}$$

(d) AB^T

$$B^T = \begin{bmatrix} 1 & 0 \\ 2 & -1 \end{bmatrix}.$$

$$\begin{aligned} AB^T &= \begin{bmatrix} 3 & 4 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 2 & -1 \end{bmatrix} \\ &= \begin{bmatrix} 3(1)+4(2) & 3(0)+4(-1) \\ 2(1)+1(2) & 2(0)+1(-1) \end{bmatrix} \\ &= \begin{bmatrix} 11 & -4 \\ 4 & -1 \end{bmatrix}. \end{aligned}$$

(e) $\text{trace}(A)$

$$\text{trace}(A) = 3 + 1 = 4$$

(f) $\text{trace}(AB)$

$$\text{trace}(AB) = 3 + 3 = 6$$

Q2 (5 points)

Given

$$M = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}.$$

(a) Eigenvalues

$$\begin{aligned} \det(M - \lambda I) &= 0 \\ \begin{vmatrix} 2-\lambda & -1 \\ -1 & 2-\lambda \end{vmatrix} &= 0 \\ (2-\lambda)^2 - 1 &= 0 \\ \lambda^2 - 4\lambda + 3 &= 0 \\ (\lambda - 1)(\lambda - 3) &= 0. \end{aligned}$$

$$\lambda_1 = 1, \qquad \lambda_2 = 3$$

(b) Eigenvectors

For $\lambda_1 = 1$:

$$\begin{aligned} (M - I)v &= 0 \\ \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} &= \begin{bmatrix} 0 \\ 0 \end{bmatrix}. \\ x - y = 0 \quad &\Longrightarrow \quad x = y. \end{aligned}$$

$$v_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

For $\lambda_2 = 3$:

$$\begin{aligned} (M - 3I)v &= 0 \\ \begin{bmatrix} -1 & -1 \\ -1 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} &= \begin{bmatrix} 0 \\ 0 \end{bmatrix}. \\ -x - y = 0 \quad &\Longrightarrow \quad x = -y. \end{aligned}$$

$$v_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

(c) Verification

For $\lambda_1 = 1$:

$$\begin{aligned} Mv_1 &= \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \\ &= \begin{bmatrix} 1 \\ 1 \end{bmatrix} = 1v_1. \end{aligned}$$

For $\lambda_2 = 3$:

$$\begin{aligned} Mv_2 &= \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix} \\ &= \begin{bmatrix} 3 \\ -3 \end{bmatrix} = 3v_2. \end{aligned}$$

$$Mv_i = \lambda_i v_i, \qquad i = 1, 2$$

Q3 (5 points)

Prove:

$$\text{trace}(AB) = \text{trace}(BA).$$

Let $A, B \in \mathbb{R}^{n \times n}$.

$$\begin{aligned} \text{trace}(AB) &= \sum_{i=1}^{n} (AB)_{ii} \\ &= \sum_{i=1}^{n} \sum_{j=1}^{n} A_{ij} B_{ji} \\ &= \sum_{i=1}^{n} \sum_{j=1}^{n} B_{ji} A_{ij} \\ &= \sum_{j=1}^{n} \sum_{i=1}^{n} B_{ji} A_{ij} \\ &= \sum_{j=1}^{n} (BA)_{jj} \\ &= \text{trace}(BA). \end{aligned}$$

$$\text{trace}(AB) = \text{trace}(BA)$$

Q4 (5 points)

Given

$$C = \begin{bmatrix} 4 & 7 \\ 2 & 6 \end{bmatrix}.$$

(a) Determinant

$$\begin{aligned}\det(C) &= ad - bc \\ &= (4)(6) - (7)(2) \\ &= 24 - 14 \\ &= 10 \neq 0.\end{aligned}$$

Therefore, C^{-1} exists.

(b) Inverse

$$C^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

$$\begin{aligned}C^{-1} &= \frac{1}{10} \begin{bmatrix} 6 & -7 \\ -2 & 4 \end{bmatrix} \\ &= \begin{bmatrix} \frac{3}{5} & -\frac{7}{10} \\ -\frac{1}{5} & \frac{2}{5} \end{bmatrix}.\end{aligned}$$

(c) Verification

$$\begin{aligned}CC^{-1} &= \begin{bmatrix} 4 & 7 \\ 2 & 6 \end{bmatrix} \frac{1}{10} \begin{bmatrix} 6 & -7 \\ -2 & 4 \end{bmatrix} \\ &= \frac{1}{10} \begin{bmatrix} 24 - 14 & -28 + 28 \\ 12 - 12 & -14 + 24 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I.\end{aligned}$$

Part 2: Probability, MLE, and MAP Estimation (50 points)

Q1 (10 points)

Given

$$P(X = k) = c\frac{3^k}{k!}, \qquad k = 0, 1, 2, \ldots$$

(a) Normalization Constant

$$\sum_{k=0}^{\infty} P(X = k) = 1$$

$$c\sum_{k=0}^{\infty} \frac{3^k}{k!} = 1$$

$$ce^3 = 1$$

$$c = e^{-3}.$$

(b) Expectation and Variance

$$P(X = k) = e^{-3}\frac{3^k}{k!} \quad \Longrightarrow \quad X \sim \text{Poisson}(3).$$

$$E[X] = e^{-3}\sum_{k=1}^{\infty} \frac{k3^k}{k!}$$

$$= 3e^{-3}\sum_{k=1}^{\infty} \frac{3^{k-1}}{(k-1)!}$$

$$= 3e^{-3}e^3 = 3.$$

$$E[X(X-1)] = e^{-3}\sum_{k=2}^{\infty} \frac{k(k-1)3^k}{k!}$$

$$= 9e^{-3}\sum_{k=2}^{\infty} \frac{3^{k-2}}{(k-2)!} = 9,$$

$$E[X^2] = E[X(X-1)] + E[X] = 9 + 3 = 12,$$

$$\text{Var}(X) = E[X^2] - (E[X])^2 = 12 - 9 = 3.$$

$$E[X] = 3, \qquad \text{Var}(X) = 3.$$

Q2 (15 points)

Given

$$P(A) = 0.5, \quad P(B) = 0.3, \quad P(A \mid B) = 0.6.$$

(a) $P(A \cap B)$

$$\begin{aligned} P(A \cap B) &= P(A \mid B)P(B) \\ &= 0.6(0.3) = 0.18. \end{aligned}$$

(b) $P(B \mid A)$

$$\begin{aligned} P(B \mid A) &= \frac{P(A \cap B)}{P(A)} \\ &= \frac{0.18}{0.5} = 0.36. \end{aligned}$$

(c) $P(A \cup B)$

$$\begin{aligned} P(A \cup B) &= P(A) + P(B) - P(A \cap B) \\ &= 0.5 + 0.3 - 0.18 = 0.62. \end{aligned}$$

Q3 (10 points)

Given $\theta = P(H)$ and $P(T) = 1 - \theta$.

$$D = (H, H, T, H, T), \qquad n = 5, \quad n_H = 3, \quad n_T = 2.$$

(a) Maximum Likelihood Estimation (MLE)

$$\begin{aligned} L(\theta) &= P(D \mid \theta) \\ &= \theta \cdot \theta \cdot (1-\theta) \cdot \theta \cdot (1-\theta) \\ &= \theta^3 (1-\theta)^2. \end{aligned}$$

$$\begin{aligned} \ell(\theta) &= \log L(\theta) \\ &= 3 \log \theta + 2 \log(1-\theta), \\ \frac{d\ell}{d\theta} &= \frac{3}{\theta} - \frac{2}{1-\theta} = 0, \\ 3(1-\theta) &= 2\theta, \\ 3 - 3\theta &= 2\theta, \\ 5\theta &= 3, \\ \hat{\theta}_{\text{MLE}} &= \frac{3}{5} = 0.6. \end{aligned}$$

$$\frac{d^2\ell}{d\theta^2} = -\frac{3}{\theta^2} - \frac{2}{(1-\theta)^2} < 0, \qquad 0 < \theta < 1.$$

$$\hat{\theta}_{\text{MLE}} = 0.6$$

(b) Maximum a Posteriori Estimation (MAP)

$$p(\theta) = \begin{cases} 2\theta, & 0 \le \theta \le 1, \\ 0, & \text{otherwise.} \end{cases}$$

$$\begin{aligned} p(\theta \mid D) &= \frac{p(D \mid \theta)p(\theta)}{p(D)} \\ &\propto \theta^3(1-\theta)^2(2\theta) \\ &= 2\theta^4(1-\theta)^2. \end{aligned}$$

$$\begin{aligned} \ell_{\text{MAP}}(\theta) &= \log 2 + 4\log\theta + 2\log(1-\theta) + C, \\ \frac{d\ell_{\text{MAP}}}{d\theta} &= \frac{4}{\theta} - \frac{2}{1-\theta} = 0, \\ 4(1-\theta) &= 2\theta, \\ 4 - 4\theta &= 2\theta, \\ 6\theta &= 4, \\ \hat{\theta}_{\text{MAP}} &= \frac{2}{3}. \end{aligned}$$

$$\frac{d^2\ell_{\text{MAP}}}{d\theta^2} = -\frac{4}{\theta^2} - \frac{2}{(1-\theta)^2} < 0, \qquad 0 < \theta < 1.$$

$$\hat{\theta}_{\text{MAP}=\frac{2}{3}\approx 0.6667}$$

Q4 (5 points)

Given $X \sim N(\mu, \sigma^2)$ and $(x_1, x_2, x_3, x_4) = (1, 3, 5, 7)$.

$$\begin{aligned} L(\mu, \sigma^2) &= \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x_i-\mu)^2}{2\sigma^2}\right) \\ &= (2\pi\sigma^2)^{-n/2} \exp\left(-\frac{\sum_{i=1}^{n}(x_i-\mu)^2}{2\sigma^2}\right), \\ \ell(\mu, \sigma^2) &= -\frac{n}{2}\log(2\pi) - \frac{n}{2}\log(\sigma^2) - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i-\mu)^2. \end{aligned}$$

MLE for μ

$$\begin{aligned} \frac{\partial \ell}{\partial \mu} &= \frac{1}{\sigma^2}\sum_{i=1}^{n}(x_i-\mu) = 0, \\ \hat{\mu}_{\text{MLE}} &= \frac{1}{n}\sum_{i=1}^{n} x_i \\ &= \frac{1+3+5+7}{4} = 4. \end{aligned}$$

MLE for σ^2

$$\frac{\partial \ell}{\partial \sigma^2} = -\frac{n}{2\sigma^2} + \frac{\sum_{i=1}^{n}(x_i - \mu)^2}{2(\sigma^2)^2} = 0,$$

$$\begin{aligned}\hat{\sigma}^2_{\text{MLE}} &= \frac{1}{n}\sum_{i=1}^{n}(x_i - \hat{\mu})^2 \\ &= \frac{1}{4}\left[(1-4)^2 + (3-4)^2 + (5-4)^2 + (7-4)^2\right] \\ &= \frac{1}{4}(9+1+1+9) = 5.\end{aligned}$$

$$\hat{\mu}_{\text{MLE}} = 4, \qquad \hat{\sigma}^2_{\text{MLE}} = 5$$

Q5 (5 points)

Given

$$x = 5, \qquad x \mid \mu \sim N(\mu, 4), \qquad \mu \sim N(0, 1).$$

$$\begin{aligned}p(\mu \mid x) &\propto p(x \mid \mu)p(\mu) \\ &\propto \exp\left(-\frac{(x-\mu)^2}{8}\right)\exp\left(-\frac{\mu^2}{2}\right) \\ &\propto \exp\left(-\frac{(5-\mu)^2}{8} - \frac{\mu^2}{2}\right).\end{aligned}$$

$$\begin{aligned}\ell(\mu) = \log p(\mu \mid x) &= -\frac{(5-\mu)^2}{8} - \frac{\mu^2}{2} + C, \\ \frac{d\ell}{d\mu} &= \frac{5-\mu}{4} - \mu = 0, \\ 5 - \mu - 4\mu &= 0, \\ 5 - 5\mu &= 0, \\ \hat{\mu}_{\text{MAP}} &= 1.\end{aligned}$$

$$\frac{d^2\ell}{d\mu^2} = -\frac{1}{4} - 1 = -\frac{5}{4} < 0.$$

$$\hat{\mu}_{\text{MAP}} = 1$$

Q6 (5 points)

Prove:

$$\text{Var}(X) = E[X^2] - [E(X)]^2.$$

$$\begin{aligned}\text{Var}(X) &= E[(X - E[X])^2] \\ &= E[X^2 - 2XE[X] + (E[X])^2] \\ &= E[X^2] - 2E[X]E[X] + (E[X])^2 \\ &= E[X^2] - 2(E[X])^2 + (E[X])^2 \\ &= E[X^2] - (E[X])^2.\end{aligned}$$

$$\text{Var}(X) = E[X^2] - [E(X)]^2$$

Part 3: Principal Component Analysis (30 points)

Q1: Covariance and Variance Derivation

Let

$$X = \begin{bmatrix} X_1 \\ X_2 \end{bmatrix}, \qquad \mu = E[X].$$

The covariance matrix is

$$\Sigma = E[(X - \mu)(X - \mu)^T].$$

For any fixed vector

$$u = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix},$$

the variance of the projection $u^T X$ is

$$\begin{aligned} \mathrm{Var}(u^T X) &= E\left[(u^T X - E[u^T X])^2\right] \\ &= E\left[(u^T X - u^T \mu)^2\right] \\ &= E\left[(u^T (X - \mu))^2\right] \\ &= E\left[u^T (X - \mu)(X - \mu)^T u\right] \\ &= u^T E\left[(X - \mu)(X - \mu)^T\right] u \\ &= u^T \Sigma u. \end{aligned}$$

Therefore,

$$\mathrm{Var}(u^T X) = u^T \Sigma u$$

For $\|u\|_2 = 1$, PCA maximizes

$$\max_{u^T u = 1} u^T \Sigma u.$$

Using a Lagrange multiplier,

$$\mathcal{L}(u, \lambda) = u^T \Sigma u - \lambda(u^T u - 1).$$

$$\begin{aligned} \nabla_u \mathcal{L} &= 2\Sigma u - 2\lambda u = 0 \\ \Sigma u &= \lambda u. \end{aligned}$$

Thus, the principal directions are the eigenvectors of Σ.

Q2: Gaussian Data Generation

Generate $n = 500$ samples from

$$X \sim \mathcal{N}(\mu, \Sigma),$$

where

$$\mu = \begin{bmatrix} 2 \\ -1 \end{bmatrix}, \qquad \Sigma = \begin{bmatrix} 3 & 1.2 \\ 1.2 & 1 \end{bmatrix}.$$

The covariance matrix is positive definite:

$$\det(\Sigma) = 3(1) - 1.2^2 = 1.56 > 0.$$

The theoretical correlation coefficient is

$$\begin{aligned}\rho_{12} &= \frac{\mathrm{Cov}(X_1, X_2)}{\sqrt{\mathrm{Var}(X_1)\,\mathrm{Var}(X_2)}} \\ &= \frac{1.2}{\sqrt{3(1)}} \\ &\approx 0.6928.\end{aligned}$$

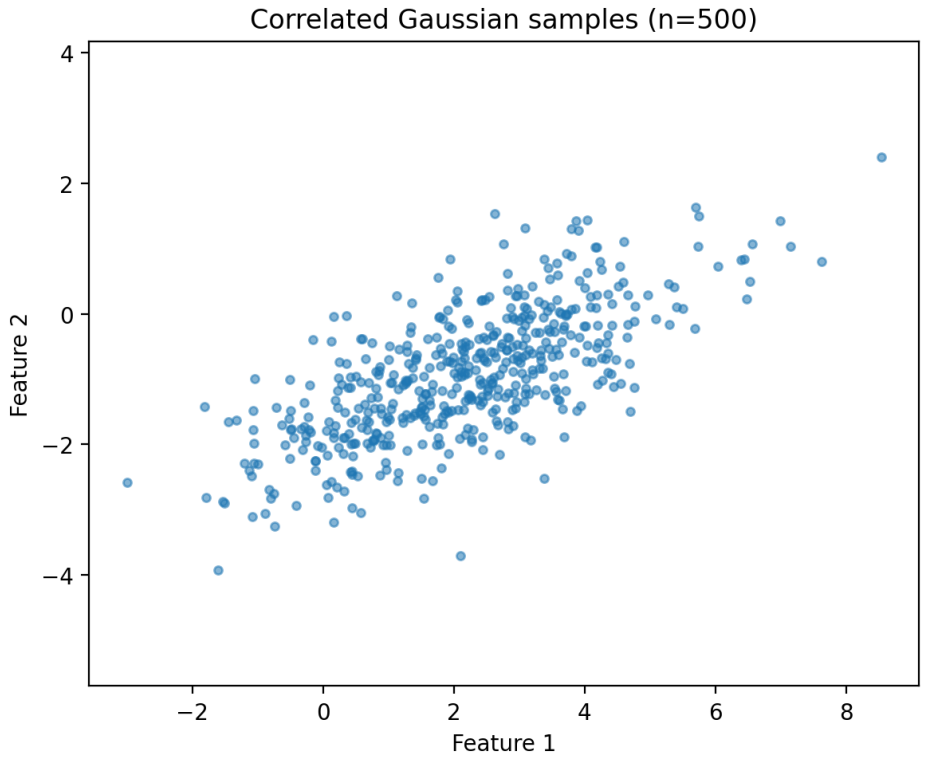


Figure 1: Correlated two-dimensional Gaussian samples.

The scatter plot shows a positively correlated distribution, consistent with the positive off-diagonal covariance $\Sigma_{12} = 1.2$.

Q3: PCA Implementation

Let the data matrix be

$$X = \begin{bmatrix} x_1^T \\ x_2^T \\ \vdots \\ x_n^T \end{bmatrix} \in \mathbb{R}^{n \times 2}.$$

(a) Mean Centering

The sample mean is

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i.$$

The centered data matrix is

$$X_c = X - \mathbf{1}_n \bar{x}^T.$$

(b) Sample Covariance Matrix

$$S = \frac{1}{n-1} X_c^T X_c.$$

For $n = 500$,

$$S = \frac{1}{499} X_c^T X_c.$$

(c) Eigendecomposition

Solve

$$Sv_i = \lambda_i v_i, \qquad i = 1, 2.$$

Since S is symmetric,

$$S = V \Lambda V^T,$$

where

$$V = \begin{bmatrix} v_1 & v_2 \end{bmatrix}, \qquad \Lambda = \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix},$$

and

$$\lambda_1 \geq \lambda_2 \geq 0, \qquad V^T V = I.$$

For the population covariance matrix,

$$\Sigma = \begin{bmatrix} 3 & 1.2 \\ 1.2 & 1 \end{bmatrix},$$

the characteristic equation is

$$\begin{aligned} \det(\Sigma - \lambda I) &= (3-\lambda)(1-\lambda) - 1.44 \\ &= \lambda^2 - 4\lambda + 1.56 = 0. \end{aligned}$$

Thus,

$$\lambda_1 = \frac{4 + \sqrt{9.76}}{2} \approx 3.562,$$
$$\lambda_2 = \frac{4 - \sqrt{9.76}}{2} \approx 0.438.$$

These are the population eigenvalues; the sample eigenvalues depend on the generated observations.

(d) PCA Projection

The transformed data is

$$Z = X_c V.$$

For dimensionality reduction to one dimension,

$$z_i = v_1^T (x_i - \bar{x}).$$

The one-dimensional projection is

$$Z_1 = X_c v_1.$$

The corresponding reconstruction is

$$\widehat{X} = Z_1 v_1^T + \mathbf{1}_n \bar{x}^T.$$

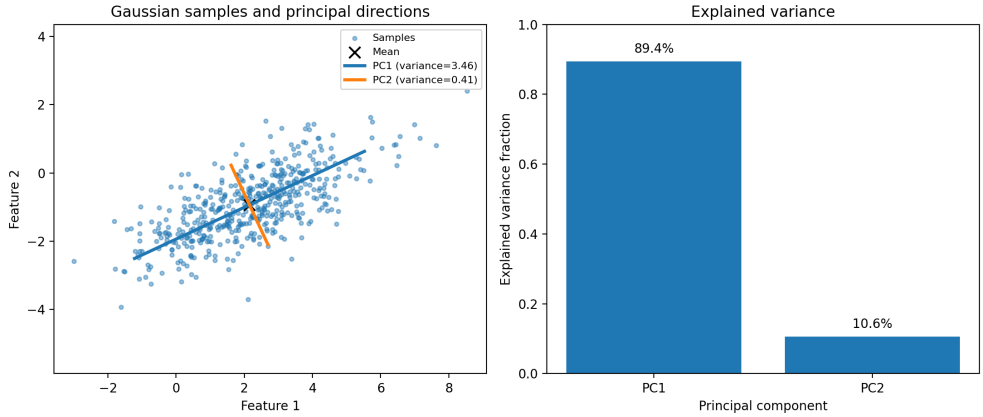


Figure 2: PCA principal directions passing through the sample mean.

The first principal component follows the direction of maximum variance. The second principal component is orthogonal to the first and captures the remaining variance.

Q4: Explained Variance

The explained variance ratio for component i is

$$r_i = \frac{\lambda_i}{\sum_{j=1}^{2} \lambda_j}.$$

Therefore,

$$r_1 = \frac{\lambda_1}{\lambda_1 + \lambda_2}, \qquad r_2 = \frac{\lambda_2}{\lambda_1 + \lambda_2}.$$

Using the population eigenvalues,

$$\begin{aligned} r_1 &= \frac{3.562}{3.562 + 0.438} \\ &\approx 0.8905, \end{aligned}$$

$$\begin{aligned} r_2 &= \frac{0.438}{3.562 + 0.438} \\ &\approx 0.1095. \end{aligned}$$

Hence,

$$r_1 \approx 89.05\%, \qquad r_2 \approx 10.95\%$$

The cumulative explained variance is

$$R_k = \sum_{i=1}^{k} r_i.$$

$$R_1 \approx 0.8905, \qquad R_2 = 1.$$

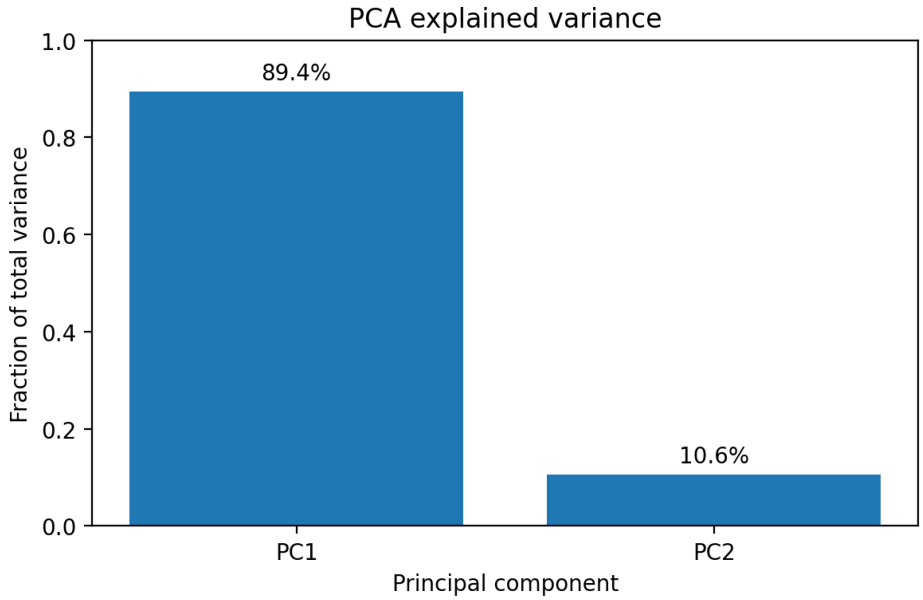


Figure 3: Explained variance ratios of the two principal components.

Retaining only the first principal component reduces the data from two dimensions to one while retaining approximately 89.05% of the population variance.

The remaining variance loss is approximately

$$1 - r_1 \approx 10.95\%.$$

For the generated sample, the exact percentages are calculated from the sample covariance eigenvalues.

The trade-off is between dimensionality reduction and information preservation: one component provides a compact representation, while two components retain all sample variance.